

IT PROJECT MANAGEMENT

TOPIC 11

PROJECT PLANNING (PROJECT RISK & PROCUREMENT MANAGEMENT)

LEARNING OBJECTIVES

a. **Understand risk and the importance** of good project risk management
b. Describes the **risk management project processes**
c. Discuss the elements of **planning risk management** and the contents of a risk management plan and common source of risk
d. Describe the process of **identifying risks** and create a risk register
e. Discuss **qualitative risk analysis** and explain how to calculate risk factors, create probability/impact matrixes, and apply the Top Ten Risk Item Tracking technique to rank risks
f. Explain **quantitative risk analysis** and how to apply decision trees, simulation, and sensitivity analysis to quantify risks
g. Provide examples of using different risk **response planning strategies** to address both negative and positive risks
h. Understand the **importance** of **project** procurement management and the increasing use of outsourcing for information technology (IT) projects
i. Describe project **procurement management processes**
j. Describe the work involved in **planning procurements** for projects, including determining the proper type of contract to use and preparing a procurement management plan, statement of work, source selection criteria, and make-or-buy analysis

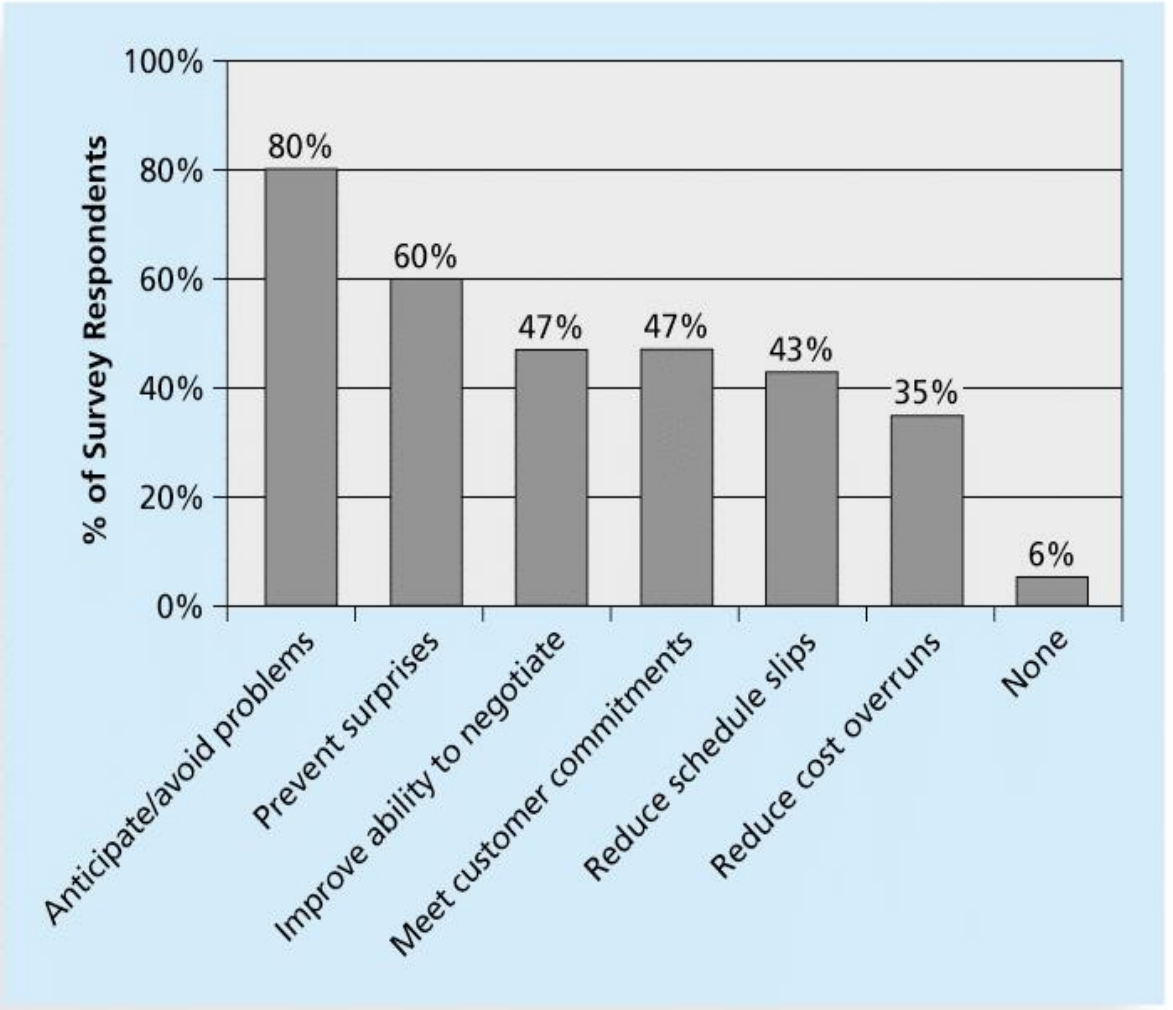
PROJECT RISK MANAGEMENT

THE IMPORTANCE OF PROJECT RISK MANAGEMENT

- Project risk management is the art and science of **identifying**, **analyzing**, and **responding** to **risk** throughout the life of a project and in the best interests of meeting project objectives
- Risk management is often overlooked in projects, but it can help improve project success by helping select good projects, determining project scope, and developing realistic estimates
- Risks are always in the future, and some can only occur at specific times in the schedule.
  - As the project progresses, **risks occur or they don't**.
  - Risks that are in the past as the project progresses are noted as being closed.
  - Risk management planning is repeated at a scheduled frequency such as at the end of a phase or a milestone.
- You can also plan to reduce procurement-related risks by making risk-related contract decisions.

BENEFITS FROM SOFTWARE RISK MANAGEMENT PRACTICES*

*Source: Kulik and Weber, KLCI Research Group



RISK

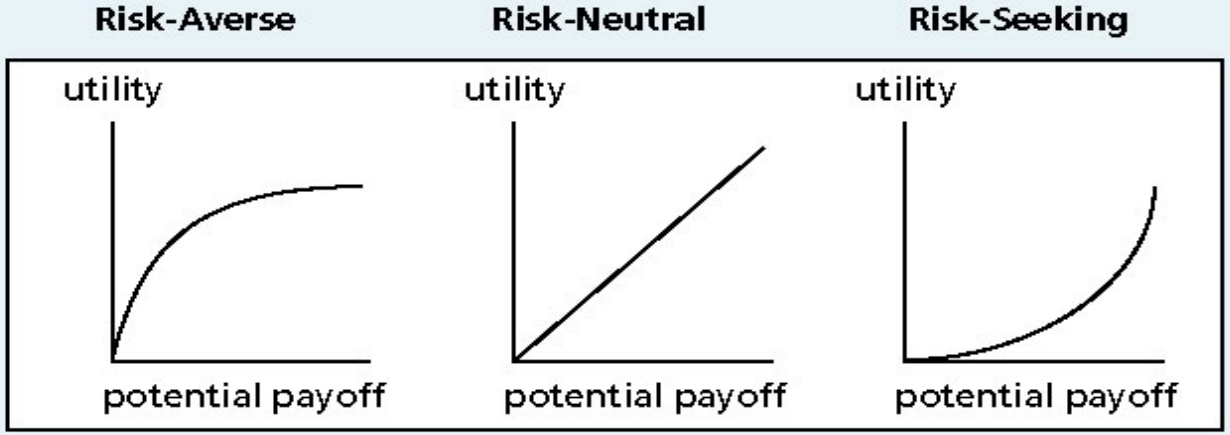
- A dictionary definition of risk is “the possibility of loss or injury”
- Negative Risk
  - Negative risk involves understanding potential problems that might occur in the project and how they might impede project success
  - Negative risk management is like a form of insurance; it is an investment
- Positive Risk
  - Positive risks are risks that result in good things happening; sometimes called **opportunities**

- A general definition of project **risk** is an uncertainty that can have a negative or positive effect on meeting project objectives

The goal of project risk management is to **minimize potential negative risks while maximizing potential positive risks**

RISK UTILITY

- **Risk utility** or **risk tolerance** is the amount of satisfaction or pleasure received from a potential payoff
  - Utility rises at a decreasing rate for people who are risk-averse
  - Those who are risk-seeking have a higher tolerance for risk and their satisfaction increases when more payoff is at stake
  - The risk-neutral approach achieves a balance between risk and payoff



PROJECT RISK MANAGEMENT PROCESSES

1. **Planning risk management** : Deciding how to **approach and plan the risk management** activities for the project
2. **Identifying risks**: Determining which **risks are likely** to **affect** a project and documenting the characteristics of each
3. **Performing qualitative risk analysis**: **Prioritizing risks** based on their probability and impact of occurrence
4. **Performing quantitative risk analysis**: Numerically **estimating the effects of risks** on project objectives
5. **Planning risk responses**: Taking steps to **enhance opportunities and reduce threats** to meeting project objectives
6. **Controlling risk**: **Monitoring identified** and residual risks, identifying new risks, carrying out risk response plans, and evaluating the effectiveness of risk strategies throughout the life of the project

PLANNING RISK MANAGEMENT

PLANNING RISK MANAGEMENT

- The main output of this process is a **risk management plan**—a plan that documents the procedures for managing risk *throughout a project*
- The project team should review project documents and understand the organization's and the sponsor's approaches to risk
- The **level of detail** will vary with the needs of the project

Topics Address in Risk Management Plan

- Methodology
- Roles and responsibilities
- Budget and schedule
- Risk categories
- Risk probability and impact
- Revised stakeholders' tolerances
- Tracking
- Risk documentation

In addition to a risk management plan, many projects also include contingency plans, fallback or backup plans, and reserves.

CONTINGENCY AND FALLBACK PLANS, CONTINGENCY RESERVES

- **Contingency plans** are predefined actions that the project team will take if an identified risk event occurs
- **Fallback plans** are developed for risks that have a high impact on meeting project objectives, and are put into effect if attempts to reduce the risk are not effective
- **Contingency reserves** or **allowances** are provisions held by the project sponsor or organization to reduce the risk of cost or schedule overruns to an acceptable level; **management reserves** are funds held for unknown risks that are NOT part of the cost baseline but ARE part of the budget and funding requirements

COMMON SOURCES OF RISK IN INFORMATION TECHNOLOGY PROJECTS

Several studies show that IT projects share some common sources of risk

The Standish Group developed an IT success potential scoring sheet based on potential risks

Success Criterion	Relative Importance
User Involvement	19
Executive management support	16
Clear Statement of Requirements	15
Proper Planning	11
Realistic Expectations	10
Smaller Project Milestones	9
Competent Staff	8
Ownership	6
Clear Visions and Objectives	3
Hard-Working, Focused Staff	3
Total	100

Other broad categories of risk help identify potential risks

- Market risk
- Financial risk
- Technology risk
- People risk
- Structure/process risk

SAMPLE RISK MANAGEMENT PLAN

Risk Management Plan – September 3

1. Methodology
The project team will review data available from the Phase I project and past training programs within Global Construction. They will also review information related to external projects similar to this one. The team will use several tools and techniques, including brainstorming, surveys, and risk-related checklists to assist in risk management.

2. Roles and Responsibilities
The project manager will be responsible for leading the team and other stakeholders in performing risk-related activities. As detailed risk-related activities and deliverables are determined, the project manager will delegate those activities to others as appropriate.

3. Funding and Timing
As specific risk-related activities and deliverables are determined, budget and schedule information will be provided. Protocols for the application of contingency and management reserves will be reviewed with the project sponsor.

4. Risk Categories
General categories and subcategories for risk on this project include business risks (suppliers and cash flow), technical risks (course content, hardware, software, and network), organizational risks (executive support, user support, supplier support, and team support), and project management risks.

5. Risk Probability and Impact
Risk probability and impact will initially be estimated as high, medium, or low based on expert judgment. High means a 75% -100% probability or impact, low means 0%-25%, and medium is in between. If more advanced scoring is needed, the project team will determine an appropriate approach.

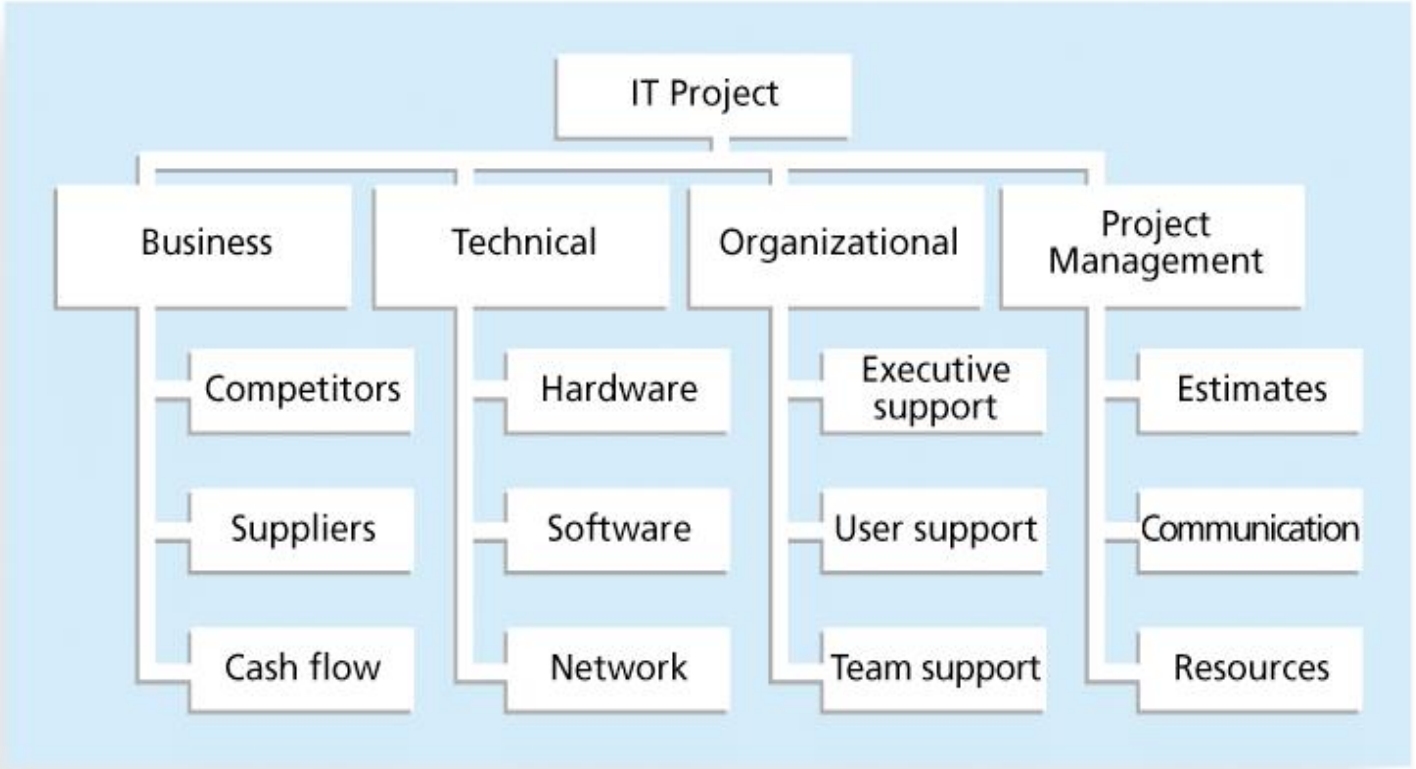
6. Stakeholder Risk Appetite
Measurable risk thresholds for each project objective will be documented for key stakeholders. These thresholds determine the acceptable overall project risk exposure and influence the probability and impact ratings.

7. Risk Documentation
All risk-related information will be summarized in a risk register. Detailed documentation will be available in a secure area on the project website.

RISK BREAKDOWN STRUCTURE

A **risk breakdown structure** is a hierarchy of potential risk categories for a project

Similar to a work breakdown structure but used to identify and categorize risks



IDENTIFYING RISK

IDENTIFYING RISKS

- You cannot manage risks until you identify them. Identifying risks involves determining which risks are likely to affect a project and documenting the characteristics of each.
- The main outputs of this process are a risk register (described later in this chapter) and a risk report
- It is important to consider risks in various categories, such as market risks, financial risks, technical risks, people risks, risks related to specific knowledge areas, and so on.
- Risk identification tools and techniques include:

Brainstorming | Review Document | Interviewing | SWOT analysis

BRAINSTORMING

- **Brainstorming** is a technique by which a group attempts to generate ideas or find a solution for a specific problem by ***amassing ideas spontaneously*** and without judgment
- An experienced facilitator should run the brainstorming session
- Be careful not to overuse or misuse brainstorming.
  - Psychology literature shows that individuals produce a greater number of ideas working alone than they do through brainstorming in small, face-to-face groups
  - **Group effects** often inhibit idea generation

INTERVIEWING AND SWOT ANALYSIS

Interviewing

- **Interviewing** is a fact-finding technique for collecting information in face-to-face, phone, e-mail, or instant-messaging discussions
- Interviewing people with similar project experience is an important tool for identifying potential risks

SWOT Analysis

- SWOT analysis (strengths, weaknesses, opportunities, and threats) can also be used during risk identification
- Helps identify the broad negative and positive risks that apply to a project

RISK REGISTER

- The main output of the risk identification process is a **list of identified risks** and other information needed to begin creating a risk register
- A **risk register** is:
  - A document that contains the results of various risk management processes and that is often displayed in a table or spreadsheet format
  - A tool for documenting potential risk events and related information
- **Risk events** refer to specific, uncertain events that may occur to the detriment or enhancement of the project
  - For example, negative risk events or threats might include the performance failure of a product produced as part of a project, delays in completing work as scheduled, increases in estimated costs, supply shortages, litigation against the company, and labor strikes.
  - Examples of positive risk events or opportunities include completing work sooner than planned or at an unexpectedly reduced cost, collaborating with suppliers to produce better products, and obtaining good publicity from the project.

RISK REGISTER CONTENTS

- An identification number for each risk event
- A rank for each risk event
- The name of each risk event
- A description of each risk event
- The category under which each risk event falls
- The root cause of each risk
- Triggers for each risk; **triggers** are indicators or symptoms of actual risk events
- Potential responses to each risk
- The **risk owner** or person who will own or take responsibility for each risk
- The probability and impact of each risk occurring.
- The status of each risk

RISK REPORT

- Overall project risk is the effect of uncertainty on the project as a whole.
- Contents of a risk report include sources of overall project risk and summary information on risk events, such as number of risks, distribution across risk categories, metrics, and trends.
- The risk report is developed progressively during the entire risk planning processes.

TABLE 11-5. SAMPLE RISK REGISTER

No.	Rank	Risk	Description	Category	Root Cause	Triggers	Potential Responses	Risk owner	Probability	Impacts	Status
R15	1										
R21	2										
R7	3										

No. : R15
Rank : 1
Risk : Poor survey response
Description : Many people dislike surveys and avoid filling them out, or if they do, they don't offer good or honest feedback
Category : Organizational/user support risk
Root cause : People don't want to take the time and think their inputs aren't important
Triggers : Low survey response rate the first few days, incomplete surveys
Risk Responses: Make sure senior management emphasizes the importance of this project and the survey for designing good courses. Have the functional managers personally mention the survey to their people and stress its importance. Offer a reward like gift certificates to the department with the most responses. Ensure that the survey instructions say it will take ten minutes or fewer to complete. Extend the deadline for survey responses.
Risk owner : Mike Sundby, Project champion
Probability : Medium
Impact : High
Status : PM will set up the meeting within the week with the project steering committee to decide which response strategies to implement

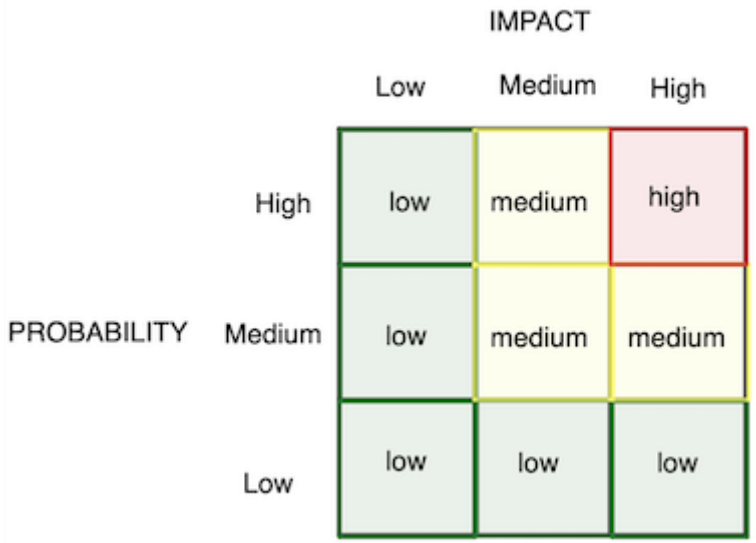
PERFORMING QUALITATIVE RISK ANALYSIS

PERFORMING QUALITATIVE RISK ANALYSIS

- Qualitative risk analysis often results in prioritizing risks as high, medium, or low.
  - A probability/impact matrix, as described in the following section, is a good technique to help decide which risks are most important on a project.
- There are two important dimensions of risk events: ***probability*** of the risk event occurring and the ***impact*** or consequence if the risk does occur.
  - People often describe a risk event probability or impact as being high, medium, or low. (High can be defined to mean a 75%-100% probability or impact, low 0%-25%, and medium in between.)
- Risk quantification tools and techniques include:
  - Probability/impact matrixes

PROBABILITY/IMPACT MATRIX

- A **probability/impact matrix** or **chart** lists the relative probability of a risk occurring on one side of a matrix or axis on a chart and the relative impact of the risk occurring on the other
- List the risks and then label each one as high, medium, or low in terms of its probability of occurrence and its impact if it did occur
- Project teams should initially focus on risk events that fall in the high sections of the probability/impact matrix and develop strategies for minimizing negative risk events and maximizing positive ones



SAMPLE OF PROBABILITY/IMPACT MATRIX

- On the Just-In-Time Training project, Kristin worked with several project stakeholders early in the project to begin identifying several negative and positive risk events related to the project.
- She held a brainstorming session in which over 100 risk events were identified.
- To differentiate between the two, she asked participants to identify negative risk events and then positive ones. After people identified a risk event and wrote it down on a sticky note, Kristin asked them to mark each one as having a high, medium, or low probability of occurrence and impact.
- They defined high to mean a 75%-100% probability, low 0%-25%, and medium in between. Kristin had posted on the whiteboard large probability/impact matrices—one for negative risk events and another for positive risk events.
- Everyone put their sticky notes in the appropriate sections of the appropriate matrix. They then examined the results to combine and reword risk events to improve collaboration and avoid duplicates.

SAMPLE OF PROBABILITY/IMPACT MATRIX

- For example, risks 1 and 2 are listed as high in both categories of probability and impact. Risks 9 and 10 are high in the probability category but low in the impact category.
- Risk 5 is high in the probability category and medium in the impact category.
- The team then discussed in detail how they planned to respond to risks, especially those in the medium and high categories, and documented the results in the risk register, as described in the following section.

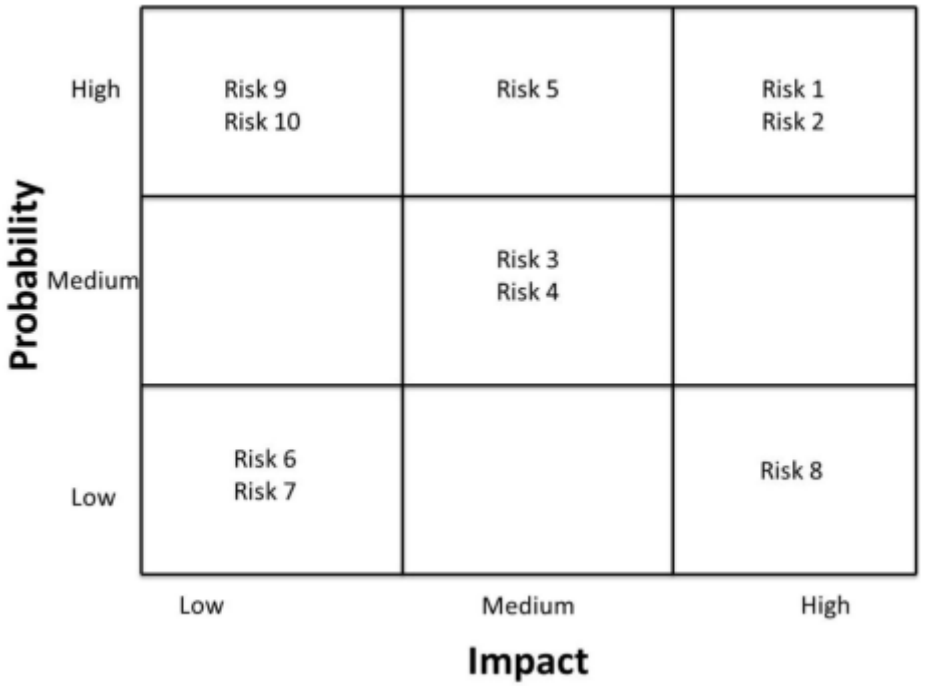
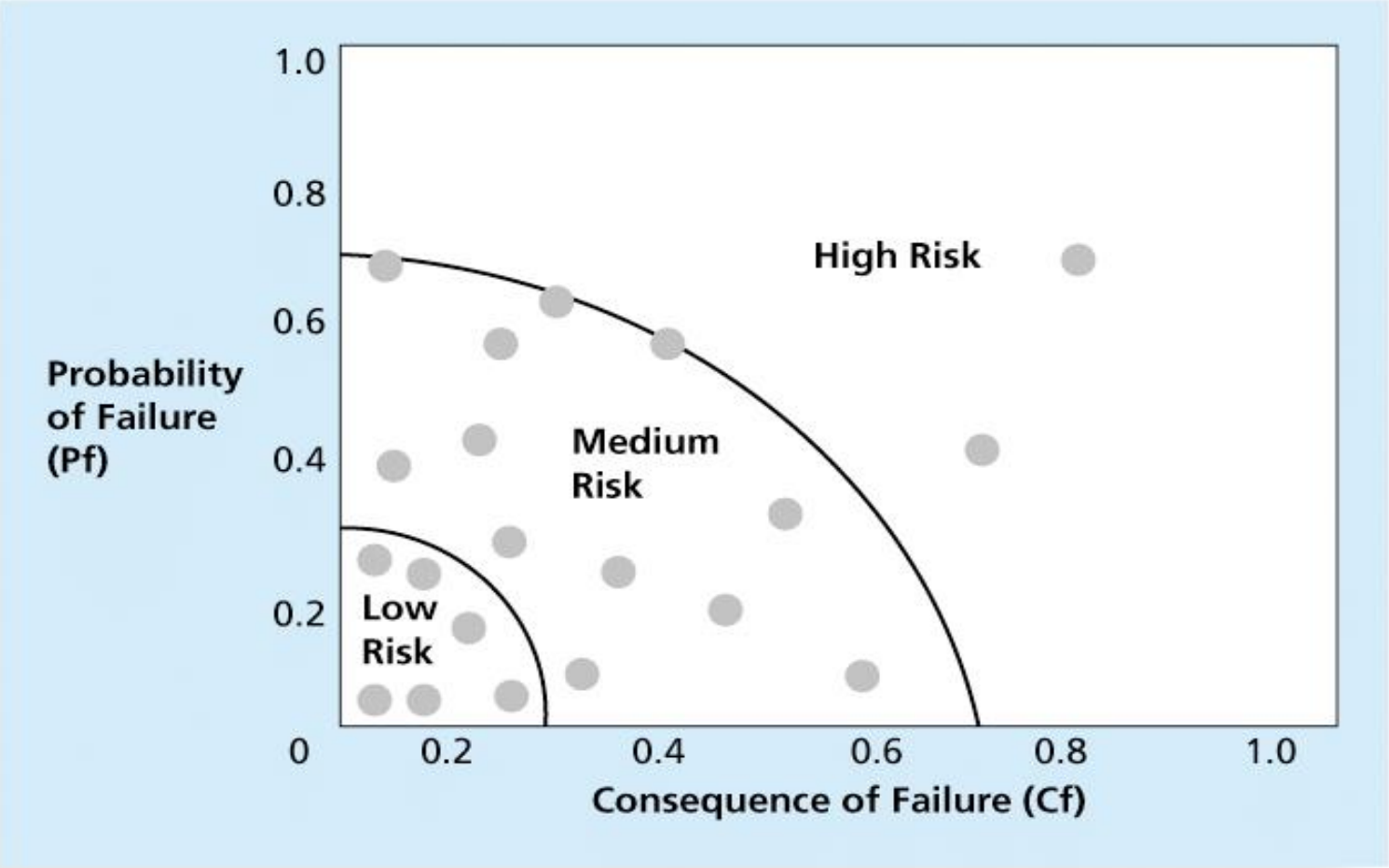


CHART SHOWING HIGH, MEDIUM, AND LOW-RISK TECHNOLOGIES



WATCH LIST

- A **watch list** is a list of risks that are low priority, but are still identified as potential risks
- Qualitative analysis can also identify risks that should be evaluated on a quantitative basis

PERFORMING QUANTITATIVE RISK ANALYSIS

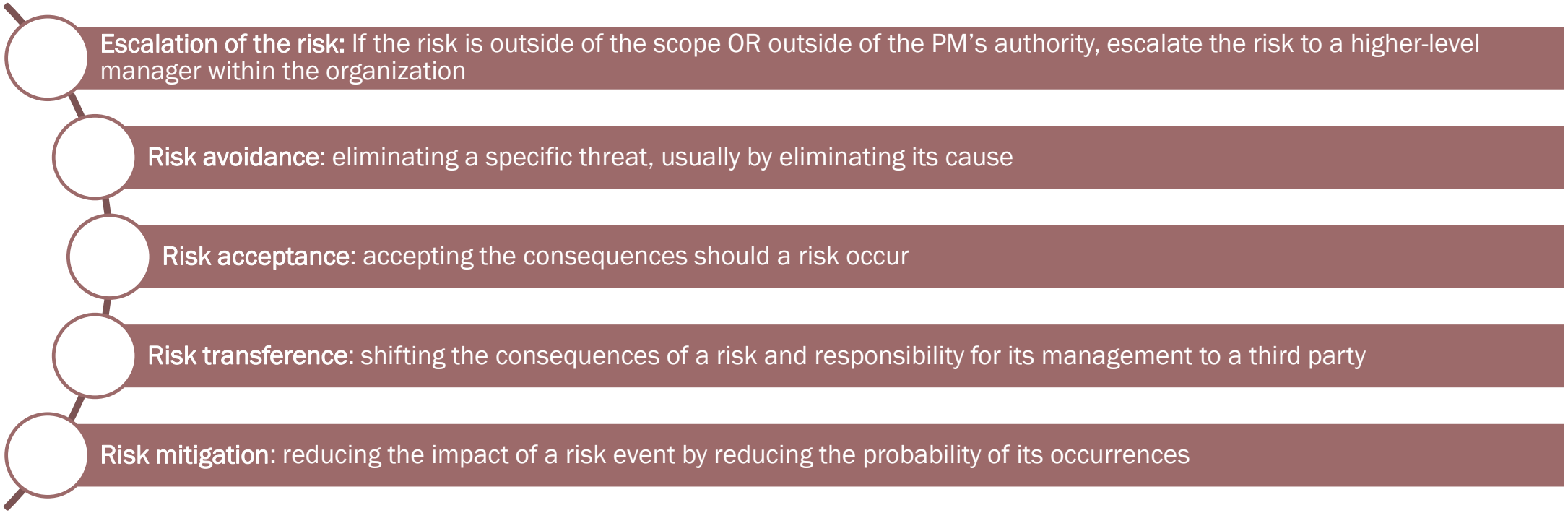
PERFORMING QUANTITATIVE RISK ANALYSIS

- On many projects, the team may *only* perform qualitative risk analysis
- Often follows qualitative risk analysis, but both can be done together
- Large, complex projects involving leading edge technologies often require extensive quantitative risk analysis
- Main techniques include:
  - Decision tree analysis
  - Simulation
  - Sensitivity analysis

PLANNING RISK RESPONSES

PLANNING RISK RESPONSES

- After identifying and quantifying risks, you must decide how to respond to them
- Four main response strategies for negative risks:



RESPONSE STRATEGIES FOR POSITIVE RISKS

- **Escalation of the risk:** If the risk is outside of the scope OR outside of the PM's authority, escalate the risk to a higher-level manager within the organization
- **Risk exploitation:** doing whatever you can to make sure the positive risk happen
- **Risk sharing :** allocating ownership of the risk to another part
- **Risk enhancement:** changing the size of the opportunity by identifying and maximizing key drivers for the positive risk
- **Risk acceptance:** the project team cannot or choose not to take any actions toward a risk.

RESIDUAL AND SECONDARY RISKS

- It’s also important to identify residual and secondary risks
- **Residual risks** are risks that remain after all of the response strategies have been implemented
- **Secondary risks** are a direct result of implementing a risk response

RISK-RELATED CONTRACT DECISIONS

- Many projects, including the Just-In-Time Training project, involve outside suppliers.
- Work done by outside suppliers or sellers should be well documented in **contracts**, which are mutually binding agreements that obligate the seller to provide the specified products or services and obligate the buyer to pay for them.
- Project managers should include clauses in contracts to help manage project risks.
  - For example, sellers can agree to be responsible for certain negative risks and incur the costs themselves if they occur. Or there can be incentive or liquidated damage clauses in contracts based on seller performance to encourage positive risks and discourage negative risks.
- You can also use certain types of contracts, such as fixedprice contracts, to reduce the risk of incurring higher costs.
- Competition for supplying goods and services can also help reduce negative risks and enhance positive risks on projects.

AGILE RISK PLANNING

- Agile projects experience uncertainties that can have negative or positive effects on meeting objectives, just like predictive projects.
- **Emphasizing value** to the customer, prioritizing work, and collaborating as a team focused on one sprint goal at a time helps to address potential risks.
- Teams should also **openly discuss impediments as part of their daily Scrum meetings**, and the Scrum master or project manager should work hard to remove impediments so the teams can focus on accomplish meaningful work.
- Many agile teams also use **risk registers,** as described earlier in this chapter.

PROJECT PROCUREMENT MANAGEMENT

IMPORTANCE OF PROJECT PROCUREMENT MANAGEMENT

- Project procurement management includes **acquiring** or **procuring** goods and services for a project from outside the organization.
- Many project managers realize the advantages of buying goods and services required for their projects, especially as sellers with better goods and services continue to become increasingly available.
  - They also realize that they can find qualified sellers throughout the world. Remember that project managers strive to do what is best for the project and the organization, and that often means acquiring goods and services from the outside.
- Good procurement management often provides a win-win situation for both buyers and sellers.
- Planning project procurement management involves **documenting procurement decisions, specifying the procurement approach, and identifying potential sellers.**
- Key outputs include procurement management plans, procurement strategies, bid documents, procurement statements of work, source selection criteria, make-or-buy decisions, independent cost estimates, change requests, project documents updates, and organizational process assets updates. Several of these outputs are described in this section.

MAKE-OR-BUY DECISIONS

- With a make-or-buy decision, an organization decides if it would benefit more by making a product or performing a service itself, or by buying the product or service from a supplier.
- If there is no need to buy products or services from outside the organization, the organization can avoid the costs involved in managing procurement management processes.
- **Make-or-buy analysis** involves estimating the internal costs of providing a product or service and comparing that estimate to the cost of outsourcing.
- Many organizations also use a lease-or-buy analysis to decide if they should purchase or lease items for a particular project.

SAMPLE MAKE-OR-BUY DECISION

For example, based on information collected in the Phase I project, it was suggested that Global Construction could probably identify internal people to conduct some of the face-to-face training or provide online advice for several courses.

Internal resources would be much less expensive than external contractors.

One of the requirements for the Just-In-Time Training project was to **provide instructor-led basic project management training and facilitation for online materials**. To train 100 employees, there would be five, two-day instructor-led classes with 20 participants in each class in the first year. There would also be 500 hours of online facilitation.

One suggestion was to have current employees conduct the basic project management training—the "make" option—instead of having an outside firm provide the training—the "buy" option.

There were a couple of project management professionals in the training department with experience in teaching face-to-face and online courses. They had taken college courses in basic project management and knew of excellent books with online materials. They also knew of several companies that conducted their own in-house courses for basic project management training and then brought in an outside expert, often a qualified college professor, who would conduct advanced courses. They found that college professors or other independent consultants provided high-quality training for much less than most training firms.

SAMPLE MAKE-OR-BUY DECISION

Make-Or-Buy Analysis
September 1

Project Name: Just-In-Time Training Project
Background: Global Construction wants to train 100 employees and will consider having the employees sent to an outside course (Buy option) or provide the education using internal employees (Make option). Assuming 20 participants/class and 2 days average course length, 10 total days of training will be needed. Assume 500 hours of online facilitation will be needed as well (5 hours per participant).
Decision Being Analyzed: Project management training
Option 1: (Make): Use in-house instructors for the instructor-led basic project management training and facilitation for online materials (includes purchasing course materials)

Estimated cost per hour for in-house trainer (excludes participant time): $60
Estimated training hours: 80 (10 total days of training X 8 hours per day)
Subtotal: $4,800 (80 hours X $60 per hour)

Materials cost: $7500 ($75/participant/course, 100 participants total)

Estimated cost per hour for online facilitation (excludes employee time): $60
Estimated hours: 500
Subtotal: $30,000 (500 hours X $60 per hour)
Total: $42,300 ($4,800 + $7,500 + $30,000)

Option 2: (Buy): Outsource instructor-led basic project management training and facilitation for online materials (includes course materials)

Estimated cost for class per participant = (Estimated Cost X Number of Days) = $500 X 2 days = $1,000
Cost of instructor-led classes ($1,000 X 100 participants)
Subtotal: $100,000
Estimated cost per hour for online facilitation (excludes student time): $100
Estimated hours: 500
Subtotal: $50,000 (500 hours X $100 per hour)
Total $150,000 ($100,000 + $50,000)

Cost Difference: $150,000 - $42,300 = $107,700
Recommendation: Because we have qualified internal staff and can purchase suitable materials, we recommend Option 1, in which we conduct the basic project management training in-house.

PROCUREMENT MANAGEMENT PLANS

- A procurement management plan is a document that describes **how the procurement processes will be managed**, from developing documentation for making outside purchases or acquisitions to contract closure.
- Like other project plans, the contents of procurement management **plans vary with project needs**.
- Some projects must follow **government directives.**
- Topics included in the procurement management plan include:
  - Guidelines on types of contracts to be used in different situations
  - Standard procurement documents or templates to be used, if applicable
  - Guidelines for creating contract work breakdown structures, statements of work, and other procurement documents
  - Roles and responsibilities of the project team and related departments, such as the purchasing or legal department
  - Guidelines on using independent estimates for evaluating sellers' cost proposals
  - Suggestions on managing multiple providers
  - Processes for coordinating procurement decisions, such as make or-buy decisions, with other project areas
  - Constraints and assumptions related to purchases and acquisitions
  - Lead times for purchases and acquisitions
  - Risk-mitigation strategies for purchases and acquisitions, such as insurance contracts and bonds
  - Guidelines for identifying prequalified sellers and organizational lists of preferred sellers
  - Procurement metrics to assist in evaluating sellers and managing contracts

TYPES OF CONTRACTS

- Contract type is a key consideration in a procurement management plan.
- Different types of contracts can be used in different situations.
- Four broad categories of contracts are fixed price, or lump sum; cost reimbursable; time and material; and indefinite delivery indefinite quantity.
- A single contract can include more than one of these categories, if it makes sense for that particular procurement.
  - For example, you could have a contract with a seller that includes purchasing specific products for a fixed price or lump sum, some services that are provided on a cost-reimbursable basis, and other services that are provided on a time-and-material basis. It is important to understand and decide which approaches to use to meet project needs.

FIXED-PRICE CONTRACTS OR LUMP-SUM CONTRACTS

- Involve a **fixed total price for a well-defined product or service**.
- The buyer incurs little risk or uncertainty in this situation because the price is predetermined.
- Sellers often pad their estimates somewhat to reduce their risk, while keeping in mind that their price must still be competitive.
  - For example, using a two-week, fixed-price contract, Global Construction could hire a consultant to develop a survey to determine requirements for its new supplier management training courses under the Just-In-Time Training project.
- Fixed-price contracts generally have **well-defined deliverables** and deadlines and **may include incentives** for meeting or exceeding selected project objectives.
  - For example, the contract could include an incentive fee that would be paid for early delivery of the survey.
- A firm-fixed price (FFP) contract has the least amount of risk for the buyer, followed by a fixed-price incentive fee (FPIF) contract. You can also use a fixed-price with economic price adjustment (FP-EPA) contract to provide for upward and downward revision of the stated contract price upon the occurrence of specified contingencies. Price adjustments can be based on established prices, actual costs of labor or material, or cost indexes of labor or material.

COST-REIMBURSABLE CONTRACTS

- Involve payment to the seller for direct and indirect actual costs.
  - For example, the salaries for people working directly on a project and materials purchased for a specific project are direct costs, whereas the cost of providing a workspace for those workers, office furniture, electricity, a cafeteria, and so on, are indirect costs.
  - Indirect costs are often calculated as a percentage of direct costs.
- Cost-reimbursable contracts **often include fees** such as a profit percentage, and they can also include incentives for meeting or exceeding selected project objectives.
- Types of contracts include **cost plus award fee (CPAF), cost plus fixed fee (CPFF), and cost plus incentive fee (CPIF).**
  - For example, many contracts to build homes are cost reimbursable contracts. The buyer might expect the home to cost a certain amount, but the total cost could vary if any of the costs of individual goods or services increase or decrease.
- The buyer **reimburses the contractor for allowable costs incurred** and pays a fee or profit percentage as well.
- Buyers absorb more of the risk with cost-reimbursable contracts than they do with fixed-price contracts.
  - For example, if the cost of wood doubles, the buyer would have to absorb the additional cost.

TIME-AND-MATERIAL (T&M) CONTRACTS

- Hybrid of both fixed price and cost-reimbursable contracts.
  - For example, an independent consultant might have a contract with a company based on a fee of $100 per hour for his or her services plus a fixed price of $10,000 for providing specific materials for the project.
  - The materials fee might also be based on approved receipts for purchasing items, with a ceiling of $10,000.
  - The consultant would send an invoice to the company each week or month, listing the materials fee, the number of hours worked, and a description of the work produced.
- This type of contract is **often used for services that are needed when the work cannot be clearly specified** and total costs cannot be estimated in a contract, such as research projects.
- Many consultants prefer time-and material contracts.

INDEFINITE DELIVERY INDEFINITE QUANTITY (IDIQ) CONTRACTS

- Provide for an indefinite quantity of goods or services for a fixed time with a stated upper and lower limit.
- This type of contract helps prevent project delays and budget overruns due to uncertainty over when goods or services would be needed.

Notes.

- Organizations can also use other documents instead of contracts, such as a memorandum of agreement or understanding, a basic ordering agreement, or a service level agreement

SAMPLE PROCUREMENT MANAGEMENT PLAN

Procurement Management Plan
September 17

Project Name: Just-In-Time Training Project
Guidelines on Types of Contracts: To reduce Global Construction's risk, contracts for the Just-In-Time Training project should be fixed price as often as possible. When goods or services cannot be well defined, other contract types may be used. The representative from the contracting department assigned to this project will work with the project manager to determine the appropriate contract type for each contract developed.
Standard procurement documents or templates: Global Construction's intranet site includes many sample documents and templates for project procurement. The project team will review these documents and templates and use them when appropriate.

Guidelines for creating procurement documents: Global Construction's intranet site provides guidelines for creating many procurement documents. The Just-In-Time Training project team should review their current work breakdown structure and scope statement to provide the basis for contract work breakdown structures and statements of work.
Roles and responsibilities: The project manager is the main contact for all procurement matters directly related to the Just-InTime Training project. The representative from the contracting department assigned to this project will coordinate with other staff in the contracting and legal departments, as needed.

BID DOCUMENTS: REQUESTS FOR INFORMATION, PROPOSALS, OR QUOTES

- **Request for Proposals**: Used to solicit proposals from prospective sellers
  - A **proposal** is a document prepared by a seller when there are different approaches for meeting buyer needs
- **Requests for Quotes**: Used to solicit quotes or bids from prospective suppliers
  - A **bid**, also called a tender or quote (short for quotation), is a document prepared by sellers providing pricing for standard items that have been clearly defined by the buyer

Topics addressed in an RFP usually include the following:

- Purpose of the RFP
- Background information, describing the organization issuing the RFP and the project itself
- Basic requirements for the products and/or services being proposed
- Hardware and software environment (for technology-related proposals)
- RFP process, describing how sellers should prepare and submit their proposals
- Statement of work and schedule information
- Appendices providing more detailed information, as appropriate

SAMPLE REQUESTS FOR PROPOSAL

Request for Proposal
August 1

Project Name: Just-In-Time Training Project
RFP Name: Qualified Sellers List for Just-In-Time Training Project

Purpose of RFP

Global Construction wants to improve training in supplier management, negotiating skills, project management, and software for its employees. In the fast-paced, ever-changing construction market, effectively training employees across a globally dispersed company with different populations is a challenge. By redesigning our current training, Global Construction can reduce training costs and improve productivity. In addition to traditional instructor-led courses provided on-site, we also want to allow our employees to learn about specific topics on a just-in-time basis by having quick access to materials and expert advice. The purpose of this RFP is to hire experts to help us find qualified sellers to develop and deliver these new training courses.

Background Information

Global Construction employs 10,000 full-time employees in ten different counties and fifteen states in the U.S. We want to increase the productivity of our employees, especially in the sales, purchasing, engineering, and information technology departments. The Just-In-Time Training Project, a one-year project, began on July 2. A key part of this project is working with outside firms to develop and provide just-in-time training in supplier management, negotiating skills, project management, and software applications. See Appendix A for detailed information on the project and specific
training needs.

Basic Requirements

The basic requirements for this work include the following:
1. Develop a list of qualified sellers to develop and provide the training as described in Appendix A.
2. Provide a summary description and detailed evaluation of each seller. Provide company brochures, websites, annual reports, and other appropriate information.
3. Work with Global Construction to develop an evaluation system to evaluate each seller.
4. Provide an objective evaluation of each seller using this evaluation system.
5. Develop a list of the top five sellers for each course.
6. Provide recommendations for developing partnerships/relationships with each of the top five sellers.
7. Complete the above work no later than September 9.

RFP Process

Prospective sellers will send written proposals to Global Construction no later than August 10. To prepare your proposal, use the outline in Appendix B, and examine Appendix C for our evaluation criteria. We expect to award the contract no later than August 20.

Statement of Work and Schedule Information

See Appendix D for a statement of work. The work must be completed no later than September 9.

Appendices

PROCUREMENT STATEMENTS OF WORK

- Another important procurement document is a **procurement statement of work (SOW)**, a document that describes the goods or services to be purchased.
  - The SOW should be included with the RFP to clarify the work that needs to be performed.
- It is a type of **scope statement that describes the work in sufficient detail** to allow prospective suppliers to both determine if they are capable of providing the goods and services required and to determine an appropriate price for the work.
- A procurement SOW should be clear, concise, and as complete as possible.
  - It should describe all services required and include performance information, such as the location and timing of the work.
  - It is important to use appropriate words in a procurement SOW—for example, *must or shall* instead of *may.*
  - The procurement SOW should specify the products and services required for the project, use industry terms, and refer to industry standards. Note that a SOW is a contractual document, and as such it could have legal ramifications.

SAMPLE PROCUREMENT STATEMENT OF WORK

Procurement Statement of Work
August 1

Project Name: Just-in-Time Training Project
Contract Name: Qualified Sellers List
Scope of Work:
1. Develop a list of qualified sellers to develop and provide the training as described in Appendix A.
2. Provide a summary description and detailed evaluation of each seller. Provide company brochures, websites, annual reports, and other appropriate information.
3. Work with Global Construction to develop an evaluation system to evaluate each seller.
4. Provide an objective evaluation of each seller using this evaluation system.
5. Develop a list of the top five sellers for each course.
6. Provide recommendations for developing partnerships/relationships with each of the top five sellers.
7. Complete the above work no later than September 9.

Location of Work:
The seller can perform the work at any location. The seller must physically meet with representatives from Global Construction in our corporate office at least twice during the term of the contract.

Period of Performance:
Work is expected to start on or around August 20 and end no later than September 9. The seller will prepare a detailed schedule for all work required, including dates for deliverables and meetings. After meeting with representatives from Global Construction to review and update the schedule, the seller will agree to the schedule for this work.

Deliverables Schedule:
The seller will prepare a detailed schedule for all required work, including dates for all deliverables and meetings. After meeting with representatives from Global Construction to review and update the schedule, the seller will agree to the schedule for this work.

Applicable Standards:
The seller will use standard software to produce the required documentation for this project. Draft and final documents will be sent via e-mail.

cceptance Criteria:
The seller will work closely with the project manager, Kristin Maur, to clarify expectations and avoid problems in providing acceptable work. Kristin will provide written acceptance/non-acceptance of all deliverables.

Special Requirements:
The seller's staff assigned to work on this contract must verify appropriate education and experience. The seller will work with Global Construction to make all travel arrangements and minimize travel costs.

SOURCE SELECTION CRITERIA AND SUPPLIER EVALUATION MATRICES

- It is highly recommended that buyers use **formal supplier evaluation procedures** to help select sellers.
- In addition to reviewing their proposals or quotes, buyers should also review sellers' past performance, talk to recent customers, interview management teams, and request sample products or demos, if applicable.
- After doing a thorough evaluation, many organizations summarize criteria and evaluations using a **supplier evaluation matrix**—a type of weighted scoring model.
- Recall from Chapter 2 that a weighted scoring model provides a systematic process for selection based on numerous criteria. For example, suppliers are often evaluated on criteria related to cost, quality, technology, past performance, and management. The weights for all of the criteria must add up to 100%

SAMPLE SUPPLIER EVALUATION MATRIX

- Kristin knew her team would have to evaluate suppliers for various goods and services as part of the Just-In-Time Training project.
- Kristin and her team reviewed Global Construction's previously used supplier evaluation matrices, and then prepared a simple matrix to evaluate the suppliers for the qualified-sellers list, as shown in Figure
- This figure was created using the supplier evaluation matrix template from the companion website, and it shows results based on three suppliers.
- In this example, Supplier 2 has the highest weighted score, so that supplier would be selected.

Just-In-Time Training Project Supplier Evaluation Matrix
September 9

Criteria	Weight	Supplier 1	Supplier 2	Supplier 3
Past performance	30%	70	90	70
Cost	25%	80	75	70
Educational background	25%	70	75	70
Management Approach	20%	85	80	70
Weighted Scores		**75.5**	**80.5**	**70**

AGILE PROCUREMENT PLANNING

- Valuing customer collaboration more than contract negotiation, as stated in the Agile Manifesto, does not mean that there are no contracts on agile projects.
- It does mean, however, that the buyer and seller should work together to create valuable products, services, and results.
- It might also mean that contracts are structured differently, allowing more incentives for sellers to work in partnership with the buyer.
- For some projects, a master agreement could be used for an overall contract.
- A **master agreement** allows for some work to be added in an appendix or supplement, allowing changes to occur without impacting the overall contract.
- Some organizations have streamlined the procurement process by using **lean-agile procurement**, an agile approach for procurements where collaboration between people is a key success factor.

CHANGE MANAGEMENT

CHANGE MANAGEMENT AND APPLYING PROJECT MANAGEMENT PRINCIPLES TO PLANNING PROJECTS

- Some projects cause big changes in organizations, so project managers as well as other managers must address larger cultural issues
- Note that change management is distinct from change control.
- Change control focuses on managing changes to project objectives. Change management focuses on the impact of projects and programs on people in organizations.
- In fact, some projects include a change manager on the project team.

MANAGING CHANGE IN ORGANIZATIONS: A PRACTICE GUIDE

- PMI publishes several practice guides, including *Managing Change in Organizations: A Practice Guide* (2013).
- The purpose of the guide is to provide "**practical help to executives and managers** who are charged with the responsibility for making change happen in and through programs and projects. It sets the practices, processes, and disciplines on managing change in the context of portfolio, program, and project management, and illustrates how change management is an essential ingredient in using project management as the vehicle for delivering organizational strategy."

MANAGING CHANGE IN ORGANIZATIONS: A PRACTICE GUIDE

The practice guide suggests the following steps required to successfully move an organization from its current state to a desired future state:

1. **Formulate change:** Create the rationale for the project/change to help people understand how the future state will be better than the current state
2. **Plan change**: Identify activities to help people prepare for changes required to move from the current state to the future state. These activities should be included in various plans in related knowledge areas.
3. **Implement change**: Show the benefits of the changes and make needed improvements or adaptations as needed to ensure the project is providing the intended outcomes. During project execution and monitoring and controlling, project teams implement their plans and make adjustments to plans as needed.
4. **Manage transition**: Address activities required after the future state is achieved. For example, many projects include a transition plan and activities related to knowledge transfer.
5. **Sustain change**: Ensure the new capabilities continue.

OTHER CHANGE MANAGEMENT MODEL

ADKAR ® Model

Kotter's 8-Step Process for

Leading Change

Virginia Satir Change Model

Bridges Transition Model

APPLYING PROJECT MANAGEMENT PRINCIPLES TO CHANGE MANAGEMENT AND PLANNING PROJECTS

The following principles relate to change management as well as planning quality, resource, communications, stakeholder, risk, and procurement management for predictive, agile, and hybrid projects:

- Create a collaborative project team environment
- Focus on value
- Build quality into processes and deliverables
- Navigate complexity
- Optimize risk responses
- Enable change to achieve the envisioned future state

CHAPTER SUMMARY

- Risk management includes several planning processes: plan risk management, identify risks, perform qualitative risk analysis, perform quantitative risk analysis, and plan risk responses.
- Planning outputs related to risk management include developing a risk management plan, risk report, risk response strategies, a risk register, and risk-related contract decisions.
- Planning outputs related to procurement management include procurement management plans, procurement strategies, bid documents, procurement statements of work, source selection criteria, make-or-buy decisions, independent cost estimates, and change requests.
- It is important to address organizational change management and apply several project management principles.
- Several change models and principles are discussed.
- Managers of agile and hybrid projects also perform planning for quality, resource, communications, stakeholder, risk, and procurement management, but the emphasis is on individuals and interactions, working solutions, customer collaboration, and responding to change. Some specific agile approaches include using Definitions of Done, information radiators, Scrum events, and lean agile procurement.